MergeRL: Learning Safe Highway On-Ramp Merging with PPO

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GitHub: https://github.com/hitesh771/mergerl

***Abstract*—On-ramp merging is a frequent cause of hard braking and rear-end collisions. We formulate it as a Markov Decision Process in which an ego car on an acceleration lane controls its longitudinal acceleration and must merge into dense highway traffic driven by the Intelligent Driver Model (IDM). We implement the simulator, PPO and Double DQN from scratch, and compare them with rule-based gap-acceptance and IDM baselines. Across 3 seeds and 3,600 evaluation episodes per agent, PPO merges in 99.4% of episodes with 1 collision in 2,400 on the training densities, and keeps 70.5% success on unseen heavier traffic where the best rule-based baseline drops to 11.2%.**

I. Problem and Environment

A single-lane highway (0–700 m) is joined by an acceleration lane whose merge zone spans 200–320 m and ends in a hard barrier. The simulator is a 1-D longitudinal point-mass model written in NumPy ($\Delta t = 0.1$ s; the agent acts every 2 sub-steps, i.e. 5 Hz). Highway cars follow the IDM,

$$a = a_{\max}\Big[1 - \big(\tfrac{v}{v_0}\big)^4 - \big(\tfrac{s^*}{s}\big)^2\Big], \quad s^* = s_0 + vT + \tfrac{v\,\Delta v}{2\sqrt{a_{\max} b}},$$

with the constant-acceleration heuristic for cut-ins and braking capped at 8 m/s^2. Every car samples its driver type between *cautious* (v_0=24 m/s, T=1.8 s) and *aggressive* (v_0=33 m/s, T=1.0 s); the density ρ is sampled per episode and inflow is steady. The ego changes lane automatically once it is inside the zone and both bumper gaps exceed only 2 m. Unsafe cut-ins are therefore *possible*: the lag car may brake hard or crash, so the agent has to learn *when and at what speed* to arrive next to a good gap. An episode ends in a collision, at the ramp end, after 40 s, or 8 s after a merge (success). The env follows the Gymnasium API and is fully seeded.

II. RL Formulation

State $\mathcal{S} \subset \mathbb{R}^{13}$ (continuous, scaled to $\approx [-1, 1]$): ego speed; ego position relative to the zone; distance to the ramp end; merged flag; gap and relative speed to the highway *lead* and *lag* cars; lag-car time-to-collision (TTC, clipped to 10 s); lag-car speed; lag-car time headway; traffic density; previous acceleration (so the policy can see the jerk it causes). Gaps are relative to the ego's longitudinal position, so a negative gap means a car is alongside.

Action $\mathcal{A} = [-1, 1]$ (continuous), scaled to an acceleration in $[-3, 3]$ m/s^2. For DQN it is discretised into 7 levels $\{-3, -2, \ldots, 3\}$ m/s^2.

Reward $\mathcal{R}$ per decision step:

$$r_t = 0.05\,\tfrac{v}{v_{\max}} - 0.02\,\tfrac{|a_t - a_{t-1}|}{6} - 0.3\,\mathbf{1}[\text{TTC} < 2\,\text{s}] - 2.0\,\mathbf{1}[a_{\text{lag}} < -4] - 0.1\,\mathbf{1}[\neg\text{merged} \wedge v < 2] - 0.02$$

plus events: $+5$ on merging, $+5$ for surviving 8 s after it, -20 for a collision, -10 for reaching the ramp end unmerged. The terms reward progress, comfort, safe spacing and courtesy (not forcing the car behind to brake harder than 4 m/s^2). The standstill penalty stops the reward hack of waiting on the ramp forever, and failing costs less than crashing.

TABLE I: Mean over seeds and densities (200 episodes/density). HB = share of episodes where the lag car brakes harder than 4 m/s^2.

	Seen ($\rho \le 55$)				Unseen (65, 75)	
Agent	Succ.↑	Coll.↓	HB↓	|jerk|↓	Succ.↑	Coll.↓
PPO	**0.994**	**0.0004**	0.13	1.29	**0.705**	**0.001**
DQN	0.965	0.025	0.17	4.22	0.584	0.046
Gap accept.	0.930	0.026	0.33	**0.58**	0.112	0.020
IDM ego	0.610	0.390	**0.10**	0.67	0.143	0.820
Random	0.520	0.455	0.28	9.99	0.002	0.872

III. Algorithms and Baselines

PPO (own PyTorch implementation): Gaussian actor and value critic (2×64 tanh MLPs, state-independent $\log\sigma$), 16 parallel envs × 256 steps per update, GAE (γ=0.99, λ=0.95), clipped surrogate (ϵ=0.2), 10 epochs × 8 minibatches, advantage and running observation normalisation, gradient clipping 0.5, and linearly decaying learning rate 3×10^{-4}. We use a **density curriculum**: 10–20 veh/km for the first 20% of training, then 10–40, then 10–55. We train for 3M steps with 3 seeds. **Double DQN** (own): 2×128 ReLU Q-network, replay 200k, Huber loss, hard target update every 2k steps, ϵ decayed from 1 to 0.05 over 30%; we train for 1M steps with 3 seeds (∼5× slower). **Baselines** (comparison only): (i) *Gap acceptance*, which plans a constant acceleration so that the ego arrives at the zone together with the gap (≥10 m) whose speed best matches its own, then follows with IDM; (ii) *IDM ego*, which follows the ramp leader and merges as soon as it is allowed; (iii) *Random*. Evaluation uses scenario seeds (≥1000) disjoint from training, 200 episodes at each density $\rho \in \{10, 25, 40, 55\}$ (seen) and $\{65, 75\}$ (*unseen*).

IV. Results

Learning. PPO reaches ≈100% training success within 0.1M steps; the dip at the start of each curriculum stage recovers within ∼0.15M steps (Fig. 3a,b). DQN learns too, but it is less stable across seeds and stage changes.

Comparison (Table I, Fig. 1). On the training densities PPO has 1 collision in 2,400 episodes (DQN: 60) and forces the lag car to brake hard 2.6× less often than gap acceptance,

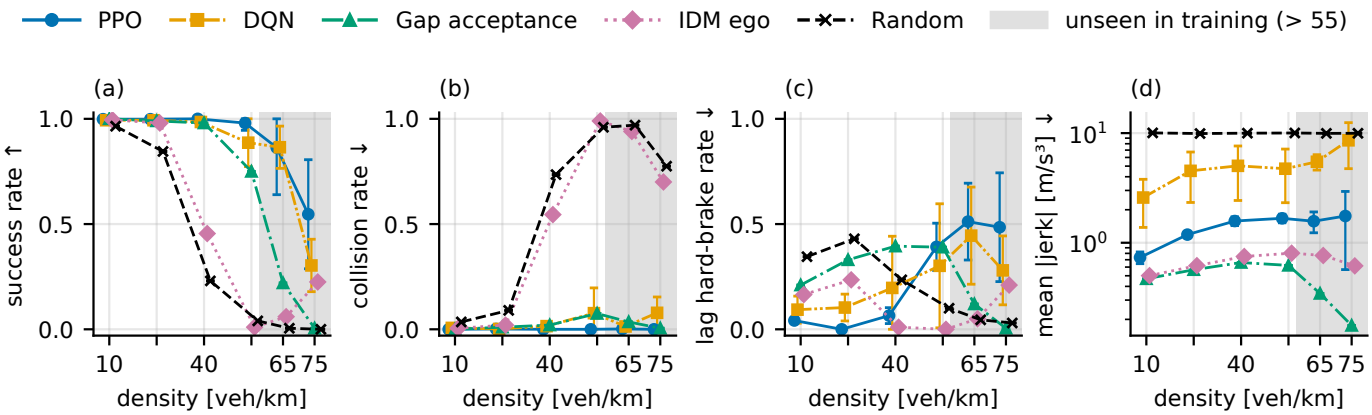


Fig. 1: Density sweep over 200 episodes per density (mean ±1 std over 3 seeds for PPO/DQN). The shaded region was never seen in training. PPO keeps success near 1 with almost no collisions and keeps the highest success in unseen traffic, at the cost of more lag-car hard braking there.

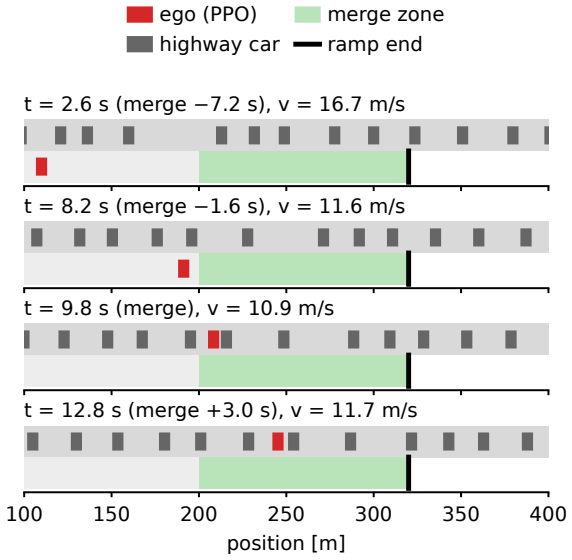


Fig. 2: A PPO episode at 40 veh/km: the agent slows down on the ramp to line up with a gap, merges at 9.8 s and then follows the traffic.

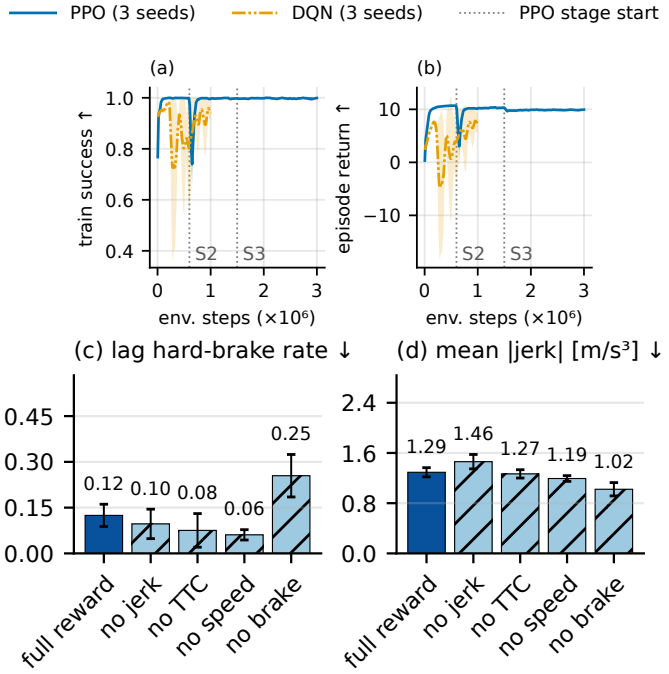


Fig. 3: (a,b) training curves (mean ±1 std over 3 seeds, 50k-step rolling mean; S2/S3 = curriculum stages). (c,d) reward ablation, PPO retrained with one term removed (3 seeds, $\rho \leq 55$).

while it merges faster (7.8 vs. 9.8 s) and drives faster (17.5 vs. 14.4 m/s). At 55 veh/km gap acceptance succeeds in only 75% of episodes, against 98% for PPO. IDM ego is fast but crashes in dense traffic, because a greedy merge rule ignores the lag car. On unseen 65–75 veh/km traffic PPO degrades gracefully (70.5% success, 0.1% collisions) and fails safely at the ramp end instead of crashing, though it forces more hard braking (HB 0.50). Gap acceptance collapses there (11.2%), because gaps of 10 m or more stop appearing. DQN's 7 coarse actions cause 3× PPO's jerk, which favours a continuous policy.

Reward ablation (Fig. 3c,d). Each term does what it was designed for. Without the brake term the lag-car hard-brake rate doubles (0.12→0.25), because the agent cuts in more aggressively. Without the jerk term |jerk| rises (1.29→1.46). Removing the TTC term gives the highest collision rate. Without the speed term the agent is 22% slower (17.5→13.6 m/s).

Limitations. PPO is still about 2× jerkier than the smooth rule-based planner. Raising the jerk weight to 0.1–0.5 did *not* reduce jerk, but did raise lag-car hard braking up to 4× in unseen traffic, so we kept 0.02. Action filtering, multi-lane traffic and learning highway drivers (multi-agent RL) are future work.

Reproducibility. `bash scripts/run_all.sh` runs the tests, then trains (seeds 0–2), evaluates and plots everything (∼2.5 h on 12 CPU cores). The trained checkpoints are in `checkpoints/`.

References

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